

DeepSeek-V2: A Strong, Economical, and Efficient MoE Language Model

236B total parameters · 21B activated per token · 128K context length

Multi-Head Latent Attention + DeepSeekMoE

The LLM Scaling Dilemma: Performance vs. Cost and Speed



The Problem

Prohibitive Training Costs

Dense models scale compute linearly with parameters. Training a 70B+ dense model demands thousands of GPUs for weeks — costs measured in millions of dollars.

KV Cache Bottleneck

Standard Multi-Head Attention stores full key-value pairs per head per token. During autoregressive generation, this cache explodes — limiting throughput.

Wasted Computation

Dense FFN layers activate all parameters for every token, regardless of relevance. Most experts are unnecessarily engaged on any given input.



The Opportunity

Sparse MoE Activation

Mixture-of-Experts architectures activate only a subset of parameters per token. Total capacity scales without proportional compute cost — train bigger, pay less.

Efficient Attention

Novel attention mechanisms can compress the KV cache dramatically without sacrificing model quality — even outperforming standard MHA in some cases.

Pareto-Optimal Scaling

By combining MoE with efficient attention, DeepSeek-V2 achieves top-tier performance at a fraction of the training and inference cost of dense counterparts.

DeepSeek-V2 Architecture: MLA + DeepSeekMoE

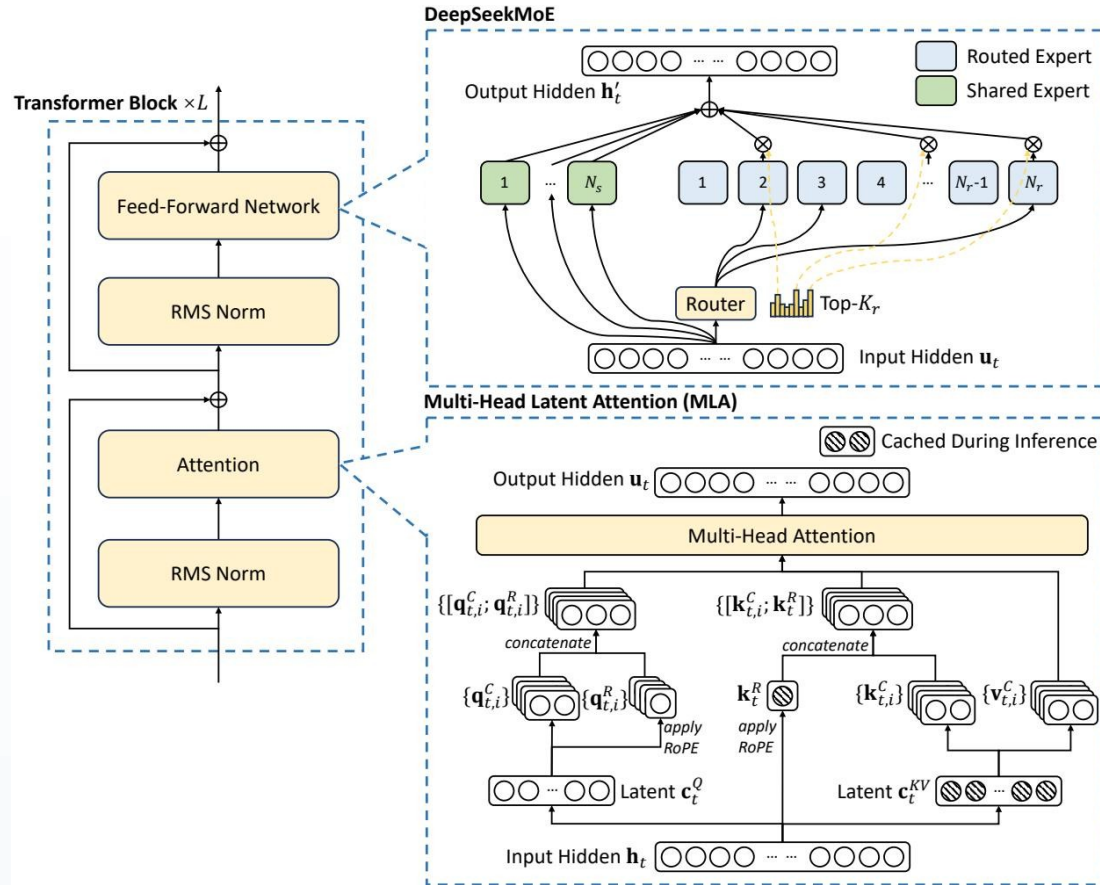


Figure: DeepSeek-V2 Transformer Block (source: arXiv:2405.04434)

Key Components

Multi-Head Latent Attention (MLA)

Replaces standard MHA with low-rank KV joint compression + decoupled RoPE.

→ **93.3% KV cache reduction**

DeepSeekMoE FFN

Shared + routed experts with fine-grained segmentation and device-limited routing.

→ **42.5% training cost reduction**

Model Scale

236B total parameters, 21B activated per token, 128K-token context window, 8.1T pretraining tokens.

→ **5.76× generation throughput**

Multi-Head Latent Attention Compresses KV Cache by 93.3%

How MLA Works

The Problem with MHA

Stores full K/V vectors per head per token.
Cache grows linearly with layers $\times$ heads $\times$ dim.

GQA / MQA Trade-off

Reduce cache by sharing keys/values across heads,
but sacrifice representational capacity and performance.

MLA Solution: Low-Rank Joint Compression

Compress K/V into a compact latent vector c_t^{KV} .
Reconstruct full K/V on-the-fly via learned projections.
Decoupled RoPE positional key preserves position info.

Result: better than MHA, 93.3% less cache.

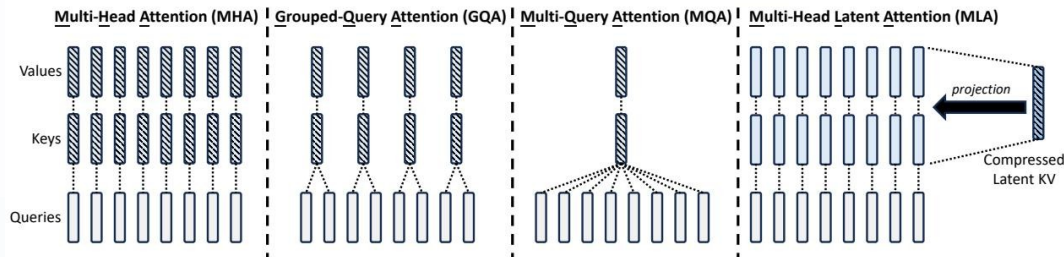


Figure: MHA vs GQA vs MQA vs MLA (source: arXiv:2405.04434)

DeepSeekMoE: Fine-Grained Experts with Shared Isolation



Expert Types

Two complementary expert classes

Shared Experts (N_s)

Always activated for every token.
Capture common, universal knowledge across all domains and tasks.

Routed Experts (N_r)

Sparsely activated via top- K_r router.
Specialize in specific knowledge domains selected per-token at runtime.

Architecture



Specialization

Fine-grained expert design

Fine-Grained Segmentation

Smaller, more numerous experts increase specialization potential. Each expert masters a narrow, precise knowledge niche.

Device-Limited Routing

Each token routes to experts on at most M devices. This bounds all-to-all communication and keeps inference latency predictable.

Efficiency



Training Stability

Balanced, robust training

Auxiliary Load-Balancing Losses

Enforce balanced load across experts and devices. Prevents expert collapse where a few experts dominate all traffic.

Token-Dropping Strategy

Drops overflow tokens from overloaded experts during training. Zero inference impact — only used to stabilize pretraining.

Stability

42.5% Training Cost Reduction and 5.76× Throughput

All metrics are relative improvements of DeepSeek-V2 vs. DeepSeek 67B (dense predecessor)



Training Cost Reduction

42.5%

vs. DeepSeek 67B

MoE sparsity + expert isolation
enables cheaper training at scale



KV Cache Reduction

93.3%

vs. DeepSeek 67B

MLA low-rank compression
replaces full K/V storage



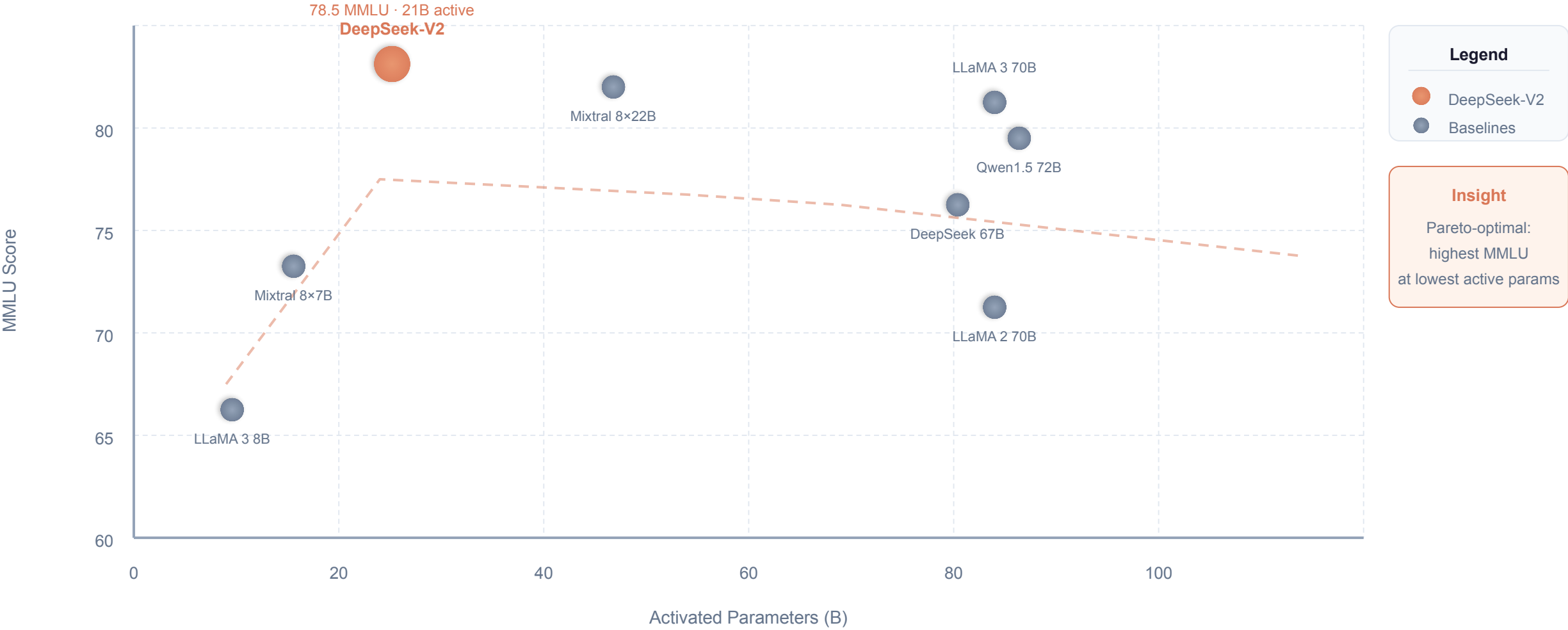
Generation Throughput

5.76×

vs. DeepSeek 67B

Smaller cache + sparse FFN
maximizes tokens per second

DeepSeek-V2 Tops MMLU at Only 21B Activated Params



DeepSeek-V2 Chat (RL) Rivals Closed-Source Models

Alignment Pipeline

Supervised Fine-Tuning (SFT)

1.5M conversational sessions for instruction following and dialogue quality.

Reinforcement Learning (GRPO)

Group Relative Policy Optimization for further alignment with human preferences.

Key Result

DeepSeek-V2 Chat (RL) outperforms **all open-source** models and **most closed-source** models on AlignBench (Chinese open-ended conversation benchmark).

Chat Evaluation Scores

Benchmark	Score
AlpacaEval 2.0 Length-controlled win rate	38.9%
MT-Bench Overall score (multi-turn)	8.97
AlignBench Overall (Chinese)	7.91

AlignBench evaluates Chinese open-ended conversation quality. DeepSeek-V2 Chat (RL) leads among open-source competitors.

KEY TAKEAWAYS

Architectural Innovation Unlocks Efficient Scaling

1

MLA is strictly better than GQA/MQA

Superior performance with 93.3% less KV cache — a new standard for efficient attention.

2

DeepSeekMoE enables economical training

Train a 236B-parameter model at 42.5% lower cost than a 67B dense model.

3

Top-tier performance at fraction of activated params

Only 21B activated parameters match or exceed 70B-class dense models on MMLU.

4

5.76× throughput makes deployment practical

Dramatic inference speedup unlocks large-scale production use cases.

5

Strongest open-source chat model in Chinese

DeepSeek-V2 Chat (RL) leads AlignBench and is competitive in English.